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PHY 615-2012
 Problem Set # 5

Problem 1: Possible values of j are
 $j = j_2 + 1, j_2, j_2 - 1$

For a given j , and eigenvalue m of J_z , the values of m_1, m_2 must be such that

$$m_1 + m_2 = m$$

Since possible values of m_1 are $1, 0, -1$ possible values of m_2 are $m-1, m, m+1$

Thus write

$$|j, m\rangle = C_1 |1, 1; j_2, m-1\rangle + C_2 |1, 0; j_2, m\rangle + C_3 |1, -1; j_2, m+1\rangle$$

Consider

$$\begin{aligned} J^2 &= \vec{J}_1^2 + \vec{J}_2^2 + 2 \vec{J}_1 \cdot \vec{J}_2 \\ &= \vec{J}_1^2 + \vec{J}_2^2 + 2 J_{1z} J_{2z} + J_{1-} J_{2+} + J_{2-} J_{1+} \end{aligned}$$

This gives

$$\begin{aligned} J^2 |j, m\rangle &= C_1 (2 + j_2(j_2+1) + 2(m-1) + J_{1-} J_{2+}) |1, 1; j_2, m-1\rangle \\ &+ C_2 (2 + j_2(j_2+1) + J_{1-} J_{2+} + J_{2-} J_{1+}) |1, 0; j_2, m\rangle \\ &+ C_3 (2 + j_2(j_2+1) - 2(m+1) + J_{1+} J_{2-}) |1, -1; j_2, m+1\rangle \end{aligned}$$

where we have used

$$J_{1+} |1, 1; j_2, m-1\rangle = J_{1-} |1, -1; j_2, m+1\rangle = 0$$

$$J_{1-} |j_1, m_1\rangle = \sqrt{j_1(j_1+1) - m_1(m_1+1)} |j_1, m_1+1\rangle$$

$$J_{1+} |j_1, m_1\rangle = \sqrt{j_1(j_1+1) - m_1(m_1-1)} |j_1, m_1-1\rangle$$

$$J_{1-} J_{2+} |1, 1; j_2, m-1\rangle$$

$$= \sqrt{2} \sqrt{j_2(j_2+1) - (m-1)m} |1, 0; j_2, m\rangle$$

$$J_{2-} J_{1+} |1, 0; j_2, m\rangle$$

$$= \sqrt{2} \sqrt{j_2(j_2+1) - m(m-1)} |1, 1; j_2, m-1\rangle$$

$$J_{2+} J_{1-} |1, 0; j_2, m\rangle$$

$$= \sqrt{2} \sqrt{j_2(j_2+1) - m(m+1)} |1, -1; j_2, m+1\rangle$$

$$J_{1+} J_{2-} |1, -1; j_2, m+1\rangle$$

$$= \sqrt{2} \sqrt{j_2(j_2+1) - (m+1)m} |1, 0; j_2, m\rangle$$

This leads to

$$J^2 |j, m\rangle$$

$$= C_1 [2 + j_2(j_2+1) + 2(m-1)] |1, 1; j_2, m-1\rangle$$

$$+ \sqrt{2} C_1 \sqrt{j_2(j_2+1) - (m-1)m} |1, 0; j_2, m\rangle$$

$$+ C_2 [2 + j_2(j_2+1)] |1, 0; j_2, m\rangle$$

$$+ \sqrt{2} C_2 \sqrt{j_2(j_2+1) - m(m+1)} |1, 1; j_2, m-1\rangle$$

$$+ \sqrt{2} C_2 \sqrt{j_2(j_2+1) - m(m+1)} |1, -1; j_2, m+1\rangle$$

$$+ C_3 [2 + j_2(j_2+1) - 2(m+1)] |1, -1; j_2, m+1\rangle$$

$$+ \sqrt{2} C_3 \sqrt{j_2(j_2+1) - (m+1)m} |1, 0; j_2, m\rangle$$

This should be

$$j(j+1) |j, m\rangle$$

This therefore requires

$$c_1 (2 + j_2(j_2+1) + 2(m-1)) + \sqrt{2} c_2 \sqrt{(j_2+m)(j_2-m+1)} = j(j+1) c_1$$

$$c_2 (2 + j_2(j_2+1)) + \sqrt{2} c_1 \sqrt{(j_2-m+1)(j_2+m)} + \sqrt{2} c_3 \sqrt{(j_2+m+1)(j_2-m)} = j(j+1) c_2$$

$$c_3 (2 + j_2(j_2+1) - 2(m+1)) + \sqrt{2} c_2 \sqrt{(j_2-m)(j_2+m+1)} = j(j+1) c_3$$

$\Rightarrow$

$$\begin{aligned} [j(j+1) - j_2(j_2+1) - 2m] c_1 \\ = \sqrt{2} (j_2+m)(j_2-m+1) c_2 \end{aligned}$$

$$\begin{aligned} [j(j+1) - j_2(j_2+1) + 2m] c_3 \\ = \sqrt{2} (j_2-m)(j_2+m+1) c_2 \end{aligned}$$

$$\begin{aligned} [j(j+1) - j_2(j_2+1) - 2] c_2 \\ = \sqrt{2} (j_2+m)(j_2-m+1) c_1 \\ + \sqrt{2} (j_2+m+1)(j_2-m) c_3 \end{aligned}$$

The first 2 equations imply 3rd automatically

Consider now $j = j_2 + 1$

$$\begin{aligned} j(j+1) - j_2(j_2+1) - 2m \\ = (j_2+1)(j_2+2) - j_2(j_2+1) - 2m \\ = 2(j_2+1) - 2m = 2[j_2 - m + 1] \end{aligned}$$

$$\begin{aligned} j(j+1) - j_2(j_2+1) + 2m \\ = 2[j_2 + m + 1] \end{aligned}$$

Then

$$\frac{c_1}{c_2} = \frac{\sqrt{2(j_2+m)(j_2-m+1)}}{2(j_2-m+1)}$$

$$= \sqrt{\frac{j_2+m}{2(j_2-m+1)}}$$

$$\frac{c_3}{c_2} = \frac{\sqrt{2(j_2-m)(j_2+m+1)}}{2(j_2+m+1)}$$

$$= \sqrt{\frac{j_2-m}{2(j_2+m+1)}}$$

This means the state vector is

$$\begin{aligned}
 |j_2+1, m\rangle &= A \left[\sqrt{\frac{j_2+m}{2(j_2-m+1)}} |1, 1; j_2, m-1\rangle \right. \\
 &\quad + |1, 0; j_2, m\rangle \\
 &\quad \left. + \sqrt{\frac{j_2-m}{2(j_2+m+1)}} |1, -1; j_2, m+1\rangle \right]
 \end{aligned}$$

The normalization constant is determined by

$$|A|^2 \left(\frac{j_2+m}{2(j_2-m+1)} + 1 + \frac{j_2-m}{2(j_2+m+1)} \right) = 1$$

$$\Rightarrow |A|^2 \frac{(2j+1)(j+1)}{(j_2-m+1)(j_2+m+1)} = 1$$

$$\Rightarrow A = \sqrt{\frac{(j_2-m+1)(j_2+m+1)}{(2j+1)(j+1)}}$$

$$\Rightarrow |j_2+1, m\rangle$$

$$\begin{aligned}
 &= \frac{1}{\sqrt{(2j+1)(j+1)}} \left\{ \frac{1}{\sqrt{2}} \sqrt{(j_2+m)(j_2+m+1)} |1, 1; j_2, m-1\rangle \right. \\
 &\quad + \sqrt{(j_2-m+1)(j_2+m+1)} |1, 0; j_2, m\rangle \\
 &\quad \left. + \frac{1}{\sqrt{2}} \sqrt{(j_2-m)(j_2-m+1)} |1, -1; j_2, m+1\rangle \right\}
 \end{aligned}$$

$$\text{For } j = j_2$$

$$\frac{C_1}{C_2} = \frac{\sqrt{2(j_2+m)(j_2-m+1)}}{-2m}$$

$$\frac{C_3}{C_2} = \frac{\sqrt{2(j_2-m)(j_2+m+1)}}{2m}$$

Thus state

$$|j_2, m\rangle$$

$$= B \left[\frac{\sqrt{(j_2+m)(j_2-m+1)}}{-\sqrt{2}m} |1, 1; j_2, m-1\rangle \right. \\ \left. + |1, 0; j_2, m\rangle \right. \\ \left. + \frac{\sqrt{(j_2-m)(j_2+m+1)}}{\sqrt{2}m} |1, -1; j_2, m+1\rangle \right]$$

The normalization determined by

$$|B|^2 \left[\frac{(j_2+m)(j_2-m+1)}{2m^2} + 1 + \frac{(j_2-m)(j_2+m+1)}{2m^2} \right] = 1$$

$$|B|^2 \frac{j_2(j_2+1)}{m^2} = 1 \Rightarrow B = \frac{m}{\sqrt{j_2(j_2+1)}}$$

$$|j_2, m\rangle$$

$$= \frac{1}{\sqrt{j_2(j_2+1)}} \left\{ -\frac{1}{\sqrt{2}} \sqrt{(j_2+m)(j_2-m+1)} |1, 1; j_2, m-1\rangle \right. \\ \left. + m |1, 0; j_2, m\rangle \right. \\ \left. + \frac{1}{\sqrt{2}} \sqrt{(j_2-m)(j_2+m+1)} |1, -1; j_2, m+1\rangle \right\}$$

For $j = j_2 - 1$

$$j(j+1) - j_2(j_2+1) \pm 2m$$

$$= -2j_2 \pm 2m$$

$\Rightarrow$

$$\frac{C_1}{C_2} = \frac{\sqrt{2(j_2+m)(j_2-m+1)}}{-2(j_2+m)}$$

$$= -\sqrt{\frac{j_2-m+1}{2(j_2+m)}}$$

$$\frac{C_3}{C_2} = -\sqrt{\frac{j_2+m+1}{2(j_2-m)}}$$

$$\Rightarrow |j_2-1, m\rangle$$

$$= C \left[\sqrt{\frac{j_2-m+1}{j_2+m}} |1, 1; j_2, m-1\rangle - |1, 0; j_2, m\rangle + \sqrt{\frac{j_2+m+1}{j_2-m}} |1, -1; j_2, m+1\rangle \right]$$

Normalization $\Rightarrow C = \sqrt{\frac{j_2^2 - m^2}{j_2(2j_2+1)}}$

$$\begin{aligned} & |j_2-1, m\rangle \\ &= \frac{1}{\sqrt{j_2(2j_2+1)}} \left[\frac{1}{\sqrt{2}} \sqrt{(j_2-m)(j_2-m+1)} |1, 1; j_2, m-1\rangle \right. \\ &\quad - \sqrt{(j_2-m)(j_2+m)} |1, 0; j_2, m\rangle \\ &\quad \left. + \frac{1}{\sqrt{2}} \sqrt{(j_2+m)(j_2+m+1)} |1, -1; j_2, m+1\rangle \right] \end{aligned}$$

Problem 3 · (16.1.2)

$$\psi(x) = (x-a)(x+a)$$

$$E(\psi) = \frac{\int dx \psi^*(x) \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \right) \psi(x)}{\int dx \psi^*(x) \psi(x)}$$

$$\int dx \psi^* \psi = \int_{-a}^a dx (x^2 - a^2)^2 = 16a^5/15$$

$$\psi'(x) = 2x \quad \psi'' = 2$$

$$E(\psi) = -\frac{\hbar^2}{m} \int_{-a}^a dx (x^2 - a^2) \cdot \frac{15}{16a^5}$$

$$= -\frac{\hbar^2}{m} \left(-\frac{4a^3}{3} \right) \left(\frac{15}{16a^5} \right)$$

$$= \frac{5\hbar^2}{4ma^2}$$

Compare

$$E_0 = \frac{\hbar^2 \pi^2}{2m(2a)^2} = \frac{\hbar^2 \pi^2}{8ma^2}$$

$$\frac{E(\psi)}{E_0} = \frac{\frac{5\hbar^2}{4ma^2}}{\frac{\hbar^2 \pi^2}{8ma^2}} = \frac{10}{\pi^2}$$

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Problem 4: (16.1.3)

Trial wave fn

$$\psi(x) = e^{-\alpha x^2}$$

$$\int dx |\psi|^2 = \sqrt{\frac{\pi}{2\alpha}}$$

$$\begin{aligned} \int dx \psi^* \left[-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} - aV_0 \delta(x) \right] \psi \\ = -\frac{\hbar^2}{2m} \int dx \psi^* \psi'' - aV_0 |\psi(0)|^2 \\ = -\frac{\hbar^2}{2m} \int dx e^{-2\alpha x^2} [2\alpha(2\alpha x^2 - 1)] - aV_0 \\ = -\frac{\hbar^2}{2m} \left(-\sqrt{\frac{\pi}{2}} \right) - aV_0 \end{aligned}$$

$$\begin{aligned} E(\psi) &= \sqrt{\frac{2\alpha}{\pi}} \left[\frac{\hbar^2}{2m} \sqrt{\frac{\pi}{2}} - aV_0 \right] \\ &= \frac{\hbar^2}{2m} \left[\alpha - \frac{2maV_0}{\hbar^2} \sqrt{\frac{2\alpha}{\pi}} \right] \end{aligned}$$

$$\frac{dE(\alpha)}{d\alpha} = \frac{\hbar^2}{2m} \left[1 - \frac{2maV_0}{\hbar^2} \sqrt{\frac{2}{\pi}} \frac{1}{2\sqrt{\alpha}} \right]$$

$$\text{Thus } \alpha_0 = \frac{(aV_0)^2}{2\pi} \cdot \left(\frac{2m}{\hbar^2} \right)^2$$

$$E(\alpha_0) = \frac{\hbar^2}{2m} \left[\left(\frac{2m}{\hbar^2} \right)^2 \frac{(aV_0)^2}{2\pi} - \frac{2maV_0}{\hbar^2} \sqrt{\frac{2}{\pi}} \frac{aV_0}{\sqrt{2\pi}} \left(\frac{2m}{\hbar^2} \right) \right]$$

$$\text{ie } E(\alpha_0) = -\frac{2m}{\hbar^2} \frac{1}{2\pi} (a v_0)^2$$

$$= -\frac{m a^2 v_0^2}{\pi \hbar^2}$$

Problem 5: (16.1.4)

$$\psi = \begin{cases} (x^2 - a^2)^2 & |x| \leq a \\ 0 & |x| > a \end{cases}$$

Then

$$\langle \psi | \psi \rangle = \int_{-a}^a dx (x^2 - a^2)^4 = \frac{256}{315} a^9$$

$$\langle \psi | \frac{\hat{p}^2}{2m} | \psi \rangle = +\frac{\hbar^2}{2m} \cdot \frac{256}{105} a^7$$

$$\langle \psi | \frac{1}{2} m \omega^2 \hat{x}^2 | \psi \rangle = \frac{1}{2} m \omega^2 \frac{256}{3465} a^{11}$$

$$\text{Thus}$$

$$\langle H \rangle = \frac{\frac{\hbar^2}{2m} \frac{256}{105} a^7 + \frac{1}{2} m \omega^2 \frac{256}{3465} a^{11}}{\frac{256}{315} a^9}$$

$$E(a) = \frac{\hbar^2}{2m} \frac{3}{a^2} + \frac{1}{2} m \omega^2 \frac{a^2}{33}$$

$$E'(a) = 0 \Rightarrow -\frac{6\hbar^2}{2ma_0^3} + \frac{m\omega^2 a_0}{33} = 0$$

This gives $a_0 = \left[\frac{\hbar^2}{m^2 \omega^2} \left(\frac{33 \times 6}{2} \right) \right]^{1/4}$

$$a_0^2 = \frac{\hbar}{m\omega} (99)^{1/2}$$

$$E(a_0) = \frac{\hbar^2}{2m} \frac{m\omega}{\hbar} \frac{1}{(99)^{1/2}} + \frac{1}{2} \frac{m\omega^2}{33} \frac{\hbar}{m\omega} (99)^{1/2}$$

$$E(a_0) = \frac{1}{2} \hbar \omega \frac{1}{\sqrt{11}} + \frac{1}{2} \hbar \omega \sqrt{11}$$

$$E(a_0) = \frac{1}{2} \hbar \omega \sqrt{\frac{12}{11}}$$